

# Zeen Zheng

Portfolio Website: [zeenz27.vercel.app](https://zeenz27.vercel.app)

425-615-3993 | [zeen6688.zz@gmail.com](mailto:zeen6688.zz@gmail.com) | [linkedin.com/in/zeenz27/](https://linkedin.com/in/zeenz27/) | [github.com/ZZeeZZ1127/](https://github.com/ZZeeZZ1127/) | U.S. Citizen

## EDUCATION

### University of Washington

Seattle, WA

*Bachelor of Science in Computer Science*

*Expected Jun 2028*

- **Relevant Coursework:** Linear Algebra, Computer Programming II/III (accelerated), System and Software Tools, Scientific Computing, Hardware/Software Interface, Foundations of Computing I & II, Data Structure and Parallelism, Algorithms, Systems Programming

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, C, C++, C#, JavaScript, TypeScript, Dart, HTML/CSS, Swift, L<sup>A</sup>T<sub>E</sub>X

**Frameworks & Libraries:** React, Next.js, Node.js, FastAPI, PySpark, Scikit-learn, pandas, NumPy, vLLM, Flutter

**Cloud & Databases:** Google Cloud Platform (GCP), AWS (Bedrock, Lambda), Supabase, Firestore

**Tools & Environments:** Git, Docker, Linux, Bash, VS Code, Unity, ROS 2, Gazebo, Figma, Vercel, Cursor, Databricks

## EXPERIENCE

### Undergraduate Research Fellow – HACR Lab

Jun 2026 – Present

*Montana State University*

*Bozeman, MT*

- Built a binary function inlining detection pipeline using **Ghidra** to disassemble  $O_3$ -compiled binaries and extract instruction-level control flow and data dependency graphs.
- Implemented dominator-based Single-Entry Single-Exit (**SESE**) region decomposition in **Python** to segment binary structures for fine-grained functional analysis.
- Deployed pre-trained Siamese Relational Graph Attention Networks (**RGATs**) to generate high-dimensional embeddings of SESE regions, detecting function inlining through cosine similarity against  $O_0$  reference set.
- Validated detection accuracy using **DWARF** debug metadata as ground truth, cross-referencing compiler-emitted records against pipeline detection to measure precision, recall, and determine cosine similarity thresholds.

### Software Engineer

Mar 2026 – Jun 2026

*SkyDrop*

*Seattle, WA*

- Built a task management dashboard with **Next.js** and **Supabase** for a 50-member team to organize the team better and deployed it using **Vercel**.
- Set up a real-time **OpenCV** and **ROS** vision pipeline for autonomous landing, detecting ArUco markers in **Gazebo** and stabilizing coordinate offsets via frame-averaging.

### AI + Backend Engineer

Oct 2025 – Jun 2026

*Husky Coding Project*

*Seattle, WA*

- Built a real-time deepfake verification for an Android application, achieving only **3-second** inference latency, by deploying the pre-trained FakeVLM vision-language model using **vLLM** on **Google Cloud Platform**.
- Set up a scalable client-server architecture capable of handling concurrent media processing requests, by designing server infrastructure and integrating **Firestore** for secure, flexible metadata storage.

## PROJECTS

**EduPath** | *AWS (Bedrock, DynamoDB, Lambda), Next.js, TypeScript, TailWind CSS*

May 2026

- Built a full-stack, AI-powered academic planning assistant utilizing **AWS Bedrock** (Claude 3.5 Sonnet) to generate personalized dynamically adaptive degree roadmaps based on user goals.
- Set up a high-performance backend infrastructure with single-table design in **AWS DynamoDB** and serverless **Lambda** functions.

**Weather-Yield Detection Model** | *PySpark, Databricks, FastAPI*

Mar 2026

- Set up a distributed **PySpark** data pipeline on **Databricks** to ingest and merge millions of rows of NOAA weather data and RMA crop yield reports.
- Trained and deployed an **XGBoost** model via **FastAPI**, using **SHAP** values and the **OpenAI Assistants API** for natural language querying of yield anomalies.

## CERTIFICATIONS

**Cisco Certified Network Associate (CCNA 200-301)**, Cisco

Jun 2025